

Write activity, factual access and control design in HyTorch

V5 results and the prospective V6 protocol

2,190

complete measurement batches

50 min 46 s

supervised execution and closure

2 batteries

exact numerical agreement

Summary. A HyTorch model trained on a synthetic factual world meets the competence criterion for common discovery facts. When writes from each of its four units are vetoed separately, factual Value loss and original-text loss respond differently. U0 produces the largest mean factual deterioration; U1 has a small, heterogeneous mean factual effect despite changing control-text loss.

The complete measurement passed its integrity and repeatability checks. However, none of the other units meets every fixed criterion for a comparable control for U0. The final result is no-control-match.

Conclusion and scope

There is evidence of a contribution to *factual access* in an already trained model. It does not yet demonstrate that a channel stops learning during training and causes hallucinations. V6 defines a new comparison between persistent and distributed blocking; its branches have not been trained.

A results and experimental-design document, not a research paper or a confirmatory claim. Includes reduced data, vector figures, reconstruction code and provenance references.

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<https://github.com/Hyphae-Research-Foundation/hytorch>

Document, figures and data: CC BY-SA 4.0. Code: Apache-2.0.

Software checks and internal review; V6 has not been empirically validated.

1 Question, system and scope

The motivating hypothesis is that a factual function could stop improving while other paths sustain a favorable aggregate loss. Studying this hypothesis requires separating three claims:

- 1. An aggregate metric can conceal performance differences between functions.
- 2. A local failure during learning causally contributes to factual errors and, under particular response conditions, to false assertions.
- 3. The mechanism is common in other models and explains a substantial share of their hallucinations.

This report informs the first question and prepares an intervention for the second. It does not estimate how often the mechanism occurs in ordinary models or its contribution to all hallucinations.

1.1 What a unit is and what is blocked

The model has four write sites, U0–U3, one per block. Proposals undergo validation, allocation and quantized application to the residual. The experimental veto converts a unit’s surviving COMMITs into explicit denials, preserving frames, original aborts and the quantization of the zero-commit path. Losing candidates are not promoted.

An active write, a nonzero gradient or a changing codebook does not establish factual knowledge by itself. Conversely, a small ablation effect can reflect redundancy or variation across facts; it does not automatically identify a unit that failed to learn.

1.2 Task and reference

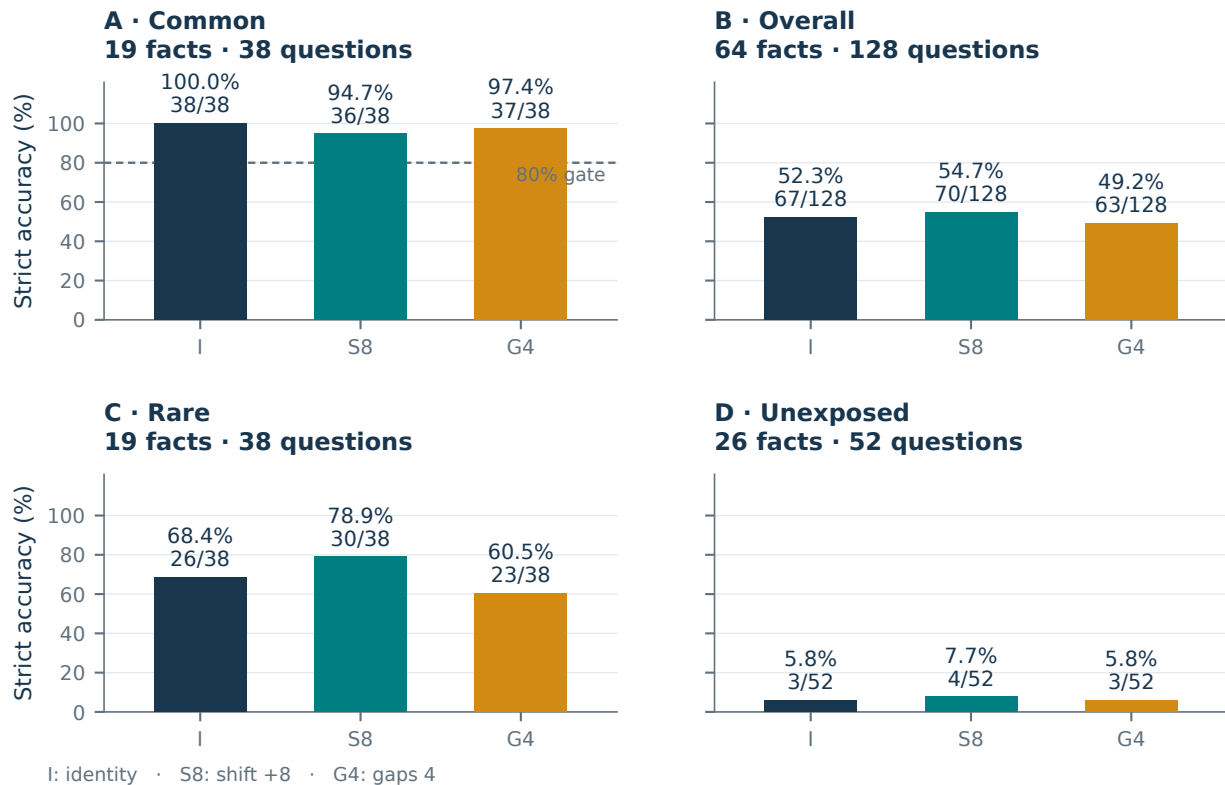
The synthetic world assigns values to entities and relations. It distinguishes common, rare and unexposed facts, alongside different entity-mention frequencies. Response truth comes from the world’s oracle: a chance-correct response to an unexposed fact is still correct.

The V5 reference undergoes auxiliary warmup W640 and 4,000 acquisition updates. It uses sequences of length 128, a vocabulary of 428 tokens and batches of 48 rows: 32 original, eight auxiliary and eight canonical. The objective combines the three losses with weights 1, 0.25, 0.25, plus global regularization weighted by 0.001. This recipe describes the evaluated reference; its loss value alone does not establish its quality.

Status	What exists at the close of this report
Observed	Reference competence and measurement of all four sites in two complete batteries.
Verified	Captures, reductions, records, closure and selection under the fixed rule.
Designed	A 17-branch V6 panel and complete schedules with mechanical tests.
Pending	V6 execution qualification, branch training and causal results.

2 Factual competence of the reference

The reference passes the *common-discovery* competence gate: at least 80% macro accuracy and strict superiority over all four declared priors in each position map. This result concerns the common-fact subset. Performance on the complete population is substantially lower.



The gate applies to common facts; overall includes all 64 facts. Each bar retains its denominator.

Figure 1: Reference accuracy by stratum and position map. Each series retains its denominator. Common-fact accuracy is not accuracy over all 128 questions. No confidence intervals are shown: maps and questions are not independent training replicates.

Map	Common	Rare	Unexposed	Overall
Identity	38/38	26/38	3/52	67/128
Shift 8	36/38	30/38	4/52	70/128
Gaps 4	37/38	23/38	3/52	63/128

The evaluation covers 64 discovery facts with two questions per fact: 19 common, 19 rare and 26 unexposed. Overall accuracies are 52.34%, 54.69% and 49.22%. Common-fact accuracies are 100%, 94.74% and 97.37%. The four common priors achieve 0%, 5.263%, 10.526% and 10.526%.

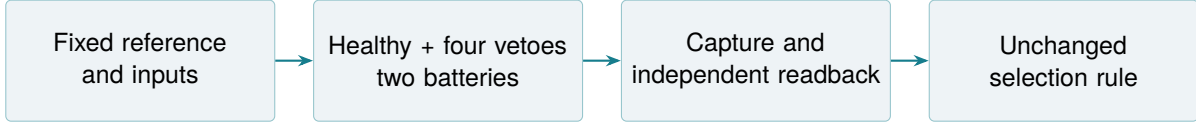
All 384 responses are present. Coverage is 100%, with no abstentions or invalid outputs; errors are false assertions in this benchmark. These results belong to the already trained reference and do not compare causal blocking or rescue training branches.

Sufficient competence within a limited scope

The reference supports a study of access contributions for common facts the system can answer. It does not establish uniform knowledge of rare or unexposed facts, or general language capability.

3 How the measurement was performed

The A4000 reference was frozen and two batteries were run with membership, order and rules fixed in advance. Each input was evaluated without a veto and with each of the four singleton vetoes. This phase involved no parameter updates or training.



3.1 Factual localization

Factual loss is measured at the public answer boundary, before supplying the correct Value token. It uses the full distribution over 428 classes, without renormalizing over the possible values:

$$\ell_{f,q,p}^{(u)} = -\log \text{softmax}(z_{f,q,p}^{(u)})[v_f], \quad (1)$$

$$\Delta_{\text{Value}}(u) = \frac{1}{|F|} \sum_{f \in F} \frac{1}{|Q_f||P|} \sum_{q \in Q_f, p \in P} \left(\ell_{f,q,p}^{(u)} - \ell_{f,q,p}^{(\emptyset)} \right). \quad (2)$$

Each fact receives equal weight, after averaging its questions and positions. A positive value means that vetoing the unit reduces the geometric mean probability assigned to the correct Value. Nats are not accuracy percentage points.

3.2 Measuring general disruption

The inventory contains all 1,120 original text occurrences for discovery entities, preserving multiplicities, the factual/mention mixture, valid targets and EOS. NLL is token-weighted within each map, and the three maps receive equal weight. The following quantities are retained for each unit:

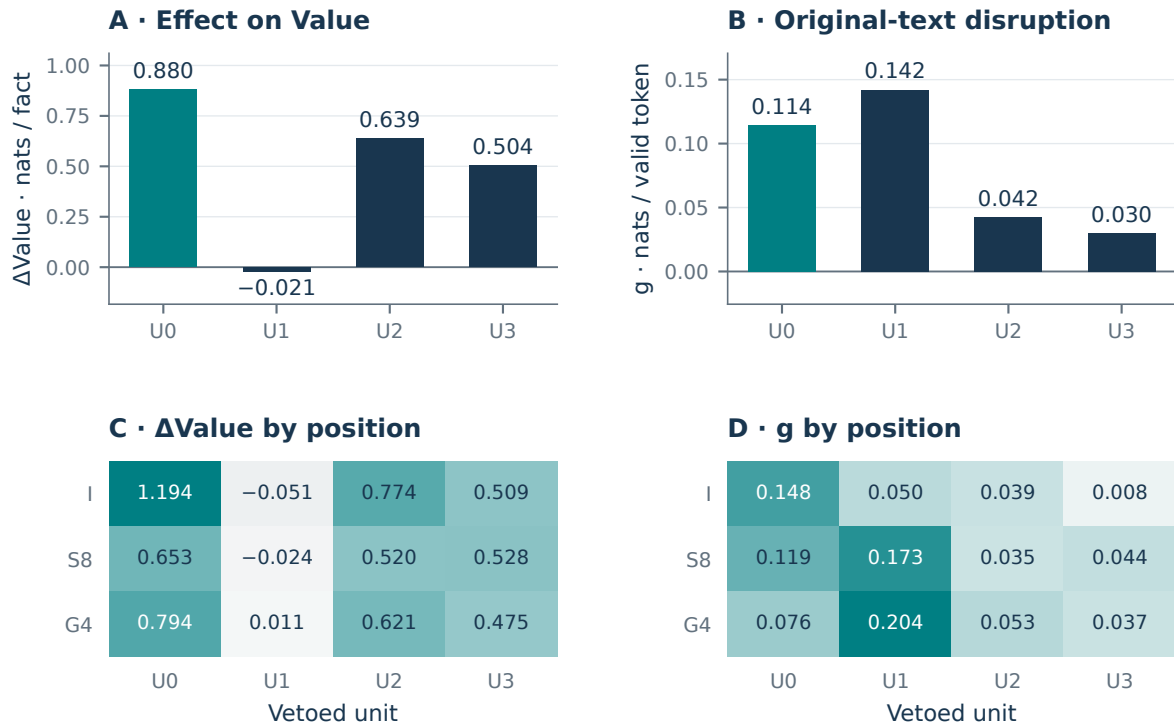
g_u	Signed change in original-text NLL: veto minus healthy.
m_u	RMS of the difference between the ordinary local write output and the quantized zero-commit output.
r_u	RMS of that local zero-commit output; $\rho_u = m_u/r_u$.
p_u	Admitted candidates among all valid candidates on the healthy path, including overflows and aborts in the denominator.

RMS values are computed from global sums of squares and counts, not by averaging batch RMS values. The proxy g uses known text from this task; it does not measure functional equivalence between units or disruption of language as a whole.

The complete population comprises 219 input sets, 2,190 batches, 34,740 real coordinates and 300 dummy rows excluded from metrics. Facts were not filtered according to the reference's correct answers.

4 Contributions of the four units

U0 leads the localizer in both batteries: $\Delta_{\text{Value}} = 0.880215$ nats, with a gap of 0.241685 over U2. U2 and U3 also have positive mean factual effects. The observed contribution is not exclusive to U0.



Δ_{Value} : equal weighting of 19 facts; g: original NLL weighted by valid tokens.

Maps have equal weight. Panels use different scales; I / S8 / G4 as in the competence figure.

Two identical batteries: one shown. They are not independent replicates; no error bars.

Figure 2: Factual and original-text effects, with their scales and position-specific breakdowns. The metrics use different populations and denominators. Both batteries produce the same values; drawing two copies would not add an independent replicate.

Unit	Δ_{Value}	g	m	r	ρ	p
U0	0.880215	0.114421	0.533184	1.178609	0.452384	0.641188
U1	-0.021274	0.142184	0.528574	1.293221	0.408727	0.637110
U2	0.638529	0.042236	0.541165	1.417010	0.381906	0.674473
U3	0.504185	0.030009	0.540632	1.529304	0.353515	0.658546

Values are rounded for readability. Both complete batteries and full-precision values are included in the report data.

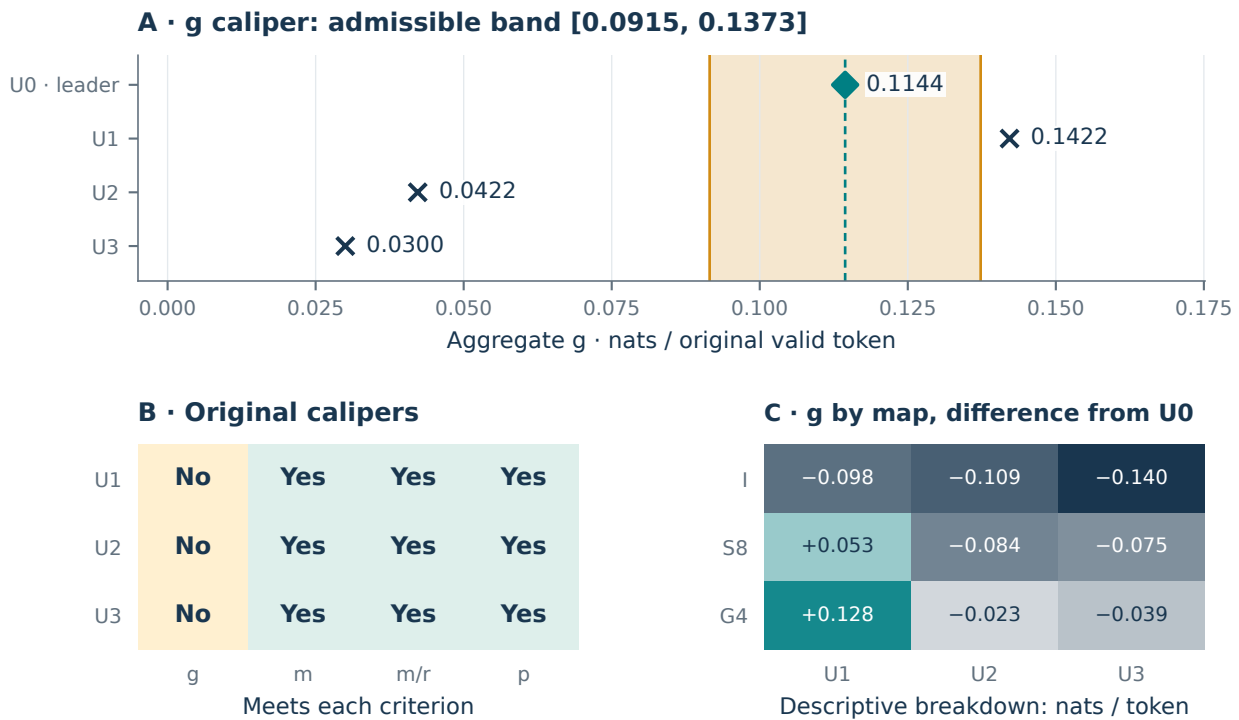
U1 has a small, slightly negative mean factual effect alongside a larger increase in original-text NLL than U0. This aggregate difference is informative but depends on the facts and maps evaluated. Its writes are active: the result does not identify a dead channel or a module dedicated exclusively to general language.

5 Why no matched control was obtained

The rule required a factual effect greater than 0.05 nats in both batteries, stable selection and a control distinct from the target satisfying four criteria. For the leading unit U0, the text-loss margin is:

$$s_g = \max(0.02, 0.20|g_{U0}|) = 0.0228841794.$$

A candidate therefore needed g within $[0.091536718, 0.137305076]$.



$s_g = 0.022884$. The band is a design threshold, not a confidence interval.

Admission requires every caliper. The map breakdown does not replace the aggregate criterion.

Both batteries produce the same matching result; no admissible control under V1.

Figure 3: Candidate evaluation under the fixed limits. The band is an admissibility criterion, not a confidence interval. All three candidates fail the text-NLL criterion and pass the other three.

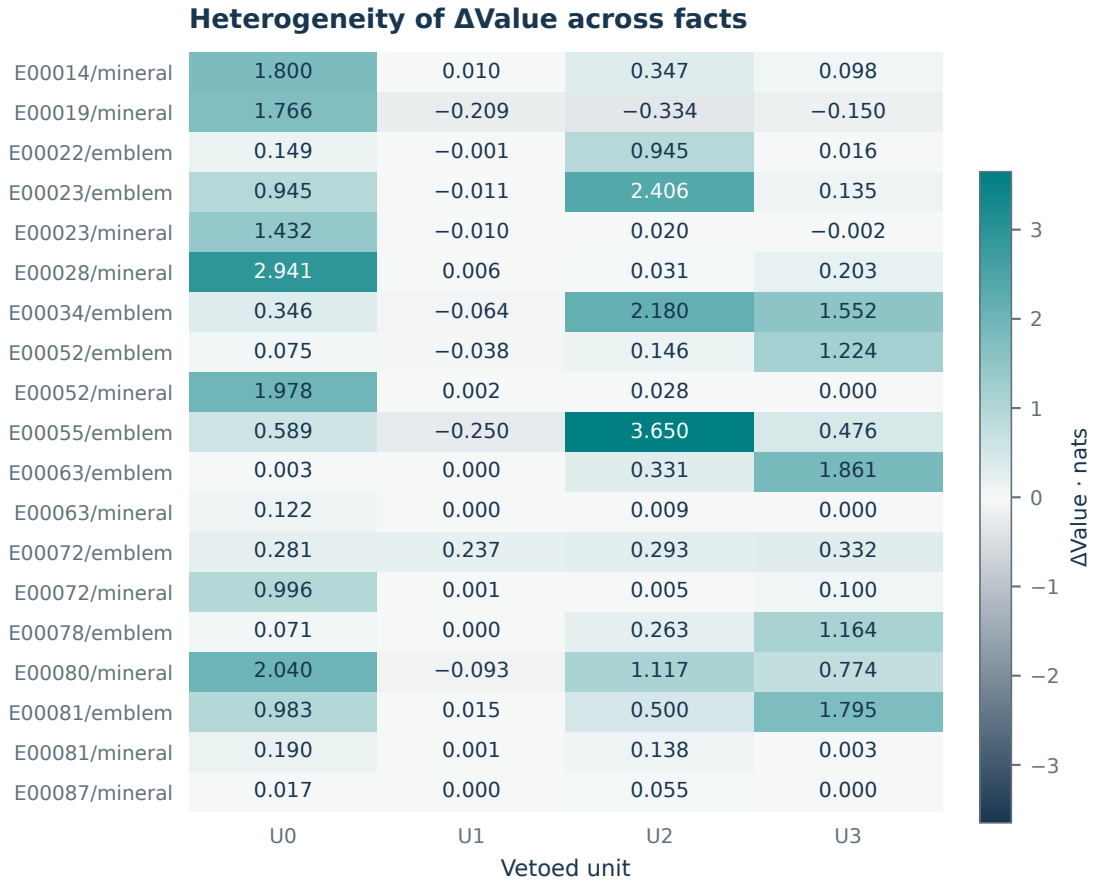
Candidate	$ g_u - g_{U0} $	m_u/m_{U0}	ρ_u/ρ_{U0}	$ p_u - p_{U0} $
U1	0.027763	0.991354	0.903495	0.004078
U2	0.072185	1.014968	0.844207	0.033285
U3	0.084411	1.013968	0.781448	0.017358
Limit	≤ 0.022884	$[2/3, 1.5]$	$[2/3, 1.5]$	≤ 0.10

U1 is the closest candidate but still exceeds the NLL limit by 0.004878687 nats. The result does not depend on rounding: the decision uses full-precision values. Calipers were not widened, the target was not replaced, and no additional batteries were added to obtain a favorable result.

Final result: no-control-match. The final target and control identifiers are null. U0 remains the localizer leader; the original matched comparison is not admitted.

6 Heterogeneity: the mean does not describe every fact

The per-fact breakdown retains information lost in a single bar. All 19 of U0's mean per-fact effects are positive, but their magnitudes vary. For U1, eight facts have positive effects and eleven have negative effects, ranging approximately from -0.249738 to 0.237074 nats.



Each cell: mean of two paraphrases $\times$ three positions. All 19 common facts are retained.

Descriptive; no CIs or p-values. E00014 = Entity_00014. Two identical batteries.

Figure 4: Factual effect by fact and unit, averaged over two questions and three maps. All 19 facts and a zero-centered scale are retained. Values are rounded: very small values may appear as zero. Cells are not independent training replicates.

Map	$\Delta_{\text{Value}}(U0)$	$\Delta_{\text{Value}}(U1)$	g_{U0}	g_{U1}
Identity	1.194326	-0.050548	0.148076	0.050147
Shift 8	0.652737	-0.024324	0.119203	0.172679
Gaps 4	0.793581	0.011050	0.075983	0.203725

U1's larger average text disruption relative to U0 does not occur in the identity map: it appears under the position transformations. The sign of U1's factual effect also varies across maps. The aggregate result therefore does not establish a simple division between a "facts" unit and a "language" unit.

This heterogeneity does not by itself explain a persistent learning failure. That hypothesis requires comparing intervened training trajectories with their controls.

7 Integrity, repeatability and interpretation limits

The measurement closed in 3,046.425 seconds, within the authorized 90 minutes, without an automatic restart. The supervisor recorded return code zero, an empty process group and confirmed closure. The final result was no longer provisional before the tables in this report were produced.

Check	Result and scope
Complete capture	2,190 batches and both batteries; no coordinates omitted to improve selection.
Independent readback	19,656,415,525 bytes read; numerical reductions and tables reconstructed from the captures.
Native records	2,193 ordered records checked: run startup and start, batch and end commitments. Query performed on a fresh copy of the stopped ledger.
Final custody	Namespace and metadata continuity for 39,722 files in 4,383 directories.
Repetition	Maximum NLL and RMS drift: zero; integer classification counts identical.
Selection	Complete rule and membership recomputed; no-control-match at closure.

7.1 What these checks establish

Raw readback checks its reductions; Native links records and receipts but does not itself determine factual truth or authenticate all neural arithmetic. The final custody check preserves continuity of previously hashed files on a trusted filesystem; it does not guarantee protection against adversarial rewrites that preserve metadata. The checks have complementary scopes.

The B16 path was qualified against the reference and sham at the same geometry. Changing from B1 to B16 is not bit-identical for every logit, although no greedy decision changed in the fixed diagnostic. Those identities and limits remain recorded; CPU qualification does not extend to GPU or XLA. The cited competence evaluation comes from a completed recovery; prior operational failures remain in the historical provenance.

7.2 What has not been demonstrated

- The two batteries repeat the same population numerically; they are not two independent seeds or worlds and do not establish statistical significance.
- Ablating a trained model measures access/prediction. It does not show that facts were never acquired or that a path stores them exclusively.
- An NLL measurement is not a classification of a generated response as a hallucination. The channel measurement did not run H/T/C/R training branches.
- An active local contribution or an active aggregate gradient does not prove factual learning by an individual unit. U1's small mean effect does not prove channel death either.

A useful finding; the causal hypothesis remains open

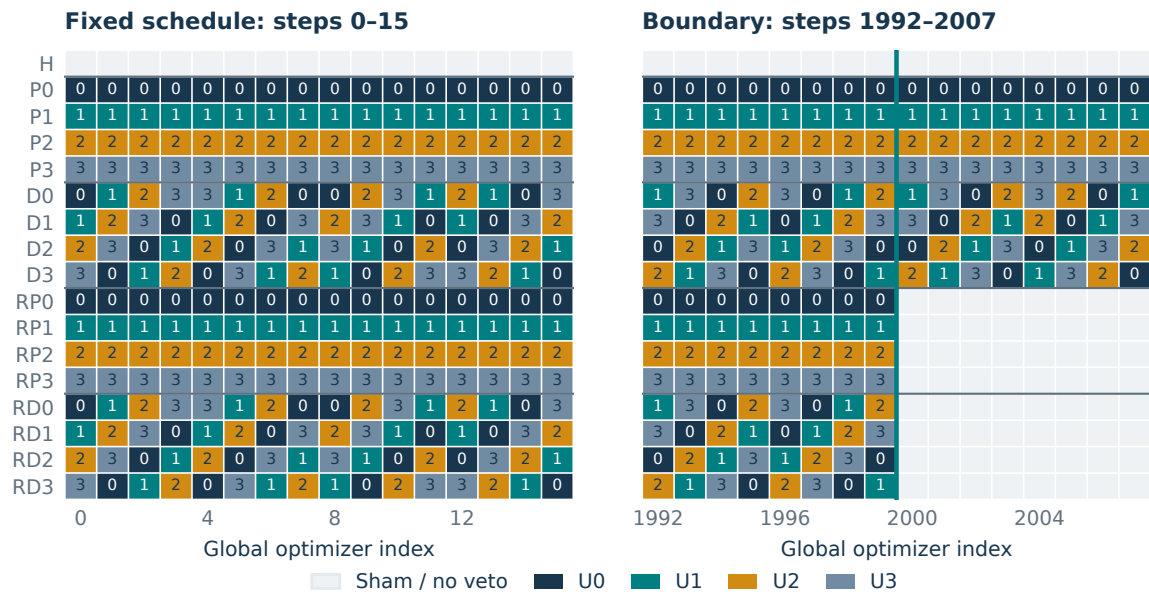
The evidence links an access intervention to measurable changes in factual probability and documents the limit of the original control. The relationship between channel death during learning and false assertions remains unproven.

8 V6: persistent and distributed blocking

The new protocol preserves no-control-match and explicitly changes the estimand. It compares persistent/concentrated blocking policies with distributed interruptions, balancing site and input marginals at every update of the panel. It does not manufacture a match for U0 or assume equal general damage along the training trajectories.

V6 · PROSPECTIVE / NOT EXECUTED

Family	Steps [0,2000)	Steps [2000,4000)
H	Empty sham	Empty sham
P0-P3	Fixed unit u	Fixed unit u
D0-D3	Distributed: $\pi_b[(k+s) \bmod 4]$	Same distributed schedule
RP0-RP3	Same prefix as P_u	Empty sham
RD0-RD3	Same prefix as D_s	Empty sham



The cell shows the vetoed unit, not the removed magnitude. Primary inference: no veto for all.

One site per forward; opportunity balance by step and block. COMMITs, RMS and g are not matched.

Figure 5: Schedules derived from the fixed V6 plan. **This is a prospective design: the branches have not been trained.** Colors identify the site scheduled for a veto; they are not measured losses or outcomes. Rescue branches remove the block from update index 2000.

The panel has 17 branches: a healthy sham H, four P branches with one fixed unit blocked, four D branches with distributed blocks and eight RP/RD rescue branches. All start from the same A0 within the world/seed and finish at A4000 with the same examples and horizon.

For $b = \lfloor t/4 \rfloor$, $k = t \bmod 4$ and a permutation π_b fixed by hashing an independent seed, without consuming model or data RNG state, D_s vetoes $\pi_b[(k+s) \bmod 4]$. Each set of four P or D branches assigns each site exactly once per update. Each D visits all sites once per four-update block; it is not permanently tied to the same auxiliary slice of the curriculum.

Each P concentrates 4,000 scheduled blocks in one site; each D distributes 1,000 per site. This balance does not equalize effective COMMITs, removed energy or gradients once parameters diverge. RP/RD match their P/D counterparts through A2000 and use sham from update index 2000, which produces A2001. Primary inference is unblocked. Each branch is executed as a separate complete run; strict resume does not permit switching schedules.

9 What V6 will estimate and what execution requires

For each final outcome Y , bars denote the mean over the four branches in each family:

$$\begin{aligned}\tau &= \bar{Y}_P - \bar{Y}_D, & \rho_P &= \bar{Y}_{RP} - \bar{Y}_P, \\ \rho_D &= \bar{Y}_{RD} - \bar{Y}_D, & \kappa &= \rho_P - \rho_D.\end{aligned}$$

τ is the total policy contrast. κ measures how that difference changes when both types of block are removed. H contextualizes competence and damage; effects at different sites need not be assumed additive.

9.1 Outcomes, data and limits

The protocol fixes a new realization of the world, exposure, model and schedule. The unit of replication is the world/seed, not the 17 branches, facts or positions. All failures are retained, and seeds are not replaced after unfavorable results.

Primary evaluation uses all common validation facts at A4000, including those H answers incorrectly; rare and unexposed strata are also reported. Factual metrics average by fact. Original-text NLL uses a separate global inventory: all original TRAIN occurrences for validation entities, weighted by valid tokens/EOS within each map. Their denominators are not mixed.

All 17 training runs close and are verified before validation capture. Public generations and full answer-boundary logits, without the correct Value in the input, are sealed first; private targets partitioned by split are opened only afterward. The historical reader that loads a combined oracle requires a qualified replacement. Validation in the old world was already evaluated in V2: it is not globally blind merely because V5 did not read it. Test remains closed in the new protocol.

A descriptive margin of 0.02 nats is fixed for jointly interpreting original-text NLL and factual deterioration by map. It is neither a functional-equivalence test nor a gate for excluding, reweighting or rerunning branches after observing them. Curves and access diagnostics are fixed in advance. A finite rescue can combine restored access with additional learning; it does not automatically identify a unique mechanism, and failure does not prove irreversibility.

9.2 Resources and technical preparation

Item	Historical serial CPU base
17 acquisition runs	39 h 2 min (approx.)
17 branches with verification, three maps and postrun	56 h 18 min 50 s
New W640 and shared initialization	+14 min 30 s (approx.)
Full pre/post example only	398.4375 GiB for 17 branches

These figures extrapolate from historical terminal records; they are not a sufficient cap for the new runner. They exclude qualification, new curves, diagnostics, other captures, copies and contingency. No parallel speedup or shared-prefix savings are assumed. GPU/AMD requires its own qualification and measured costs.

The schedule passed 62 tests and the package passed six additional checks. That validates pure design components. The new schedule family in the kernel, guard and Native, the export and private reader, qualified initialization and numerical execution, and the full budget remain pending. No V6 training has been performed.

10 Reproducibility and provenance

This report includes LaTeX source, five vector figures, reduced data and reconstruction scripts. A clean clone can rebuild the figures and recompute selection without loading a model, querying Native or downloading the complete raw captures.

Included artifact	What it supports checking
data/report-data.json	Curated competence, unit, position and fact data, limits and the illustrated schedule.
data/provenance.json	Original sources, extraction fields and hashes; originals distinguished from reduced exports.
Tables and selector in data/ design/ make_figures.py	Recompute both batteries' aggregates and decision under the original rule from their complete tables. V6 protocol, complete schedule and pure balance verification. Rebuild localized figures from portable data; no post hoc example selection.
report-en.tex, Makefile	Compile the English technical PDF; no experimental execution.

Reproducing tables and figures does not replace full revalidation from the original raw files, checkpoints and ledgers, which are not included. The historical readback counter recorded approximately 19.66 GB read; it is not the combined size of those three evidence types. The exporter authenticates the pinned source JSON and documents when available, without revalidating raw blobs or checkpoints. This package neither reconstructs the historical training trajectory nor qualifies a future run merely by containing its hashes.

10.1 Principal identifiers

Repository-relative paths and the complete source list are in the provenance manifest. The following SHA256 hashes identify cited closures and rules. The V6 protocol hash is for the frozen Spanish source, not its English translation.

Evidence	SHA256 of the original artifact
V5 competence	bdb63840ae28176e5cbf22aded90ad78585804415e0d5579ca9bb63da1bee9db
Worker closure	f223ec5ec1101a67528a4cd54d38ce9e4d7dd405ad0c26586f63c360d42810a2
Supervisor closure	18b41a1980e6eef486a1747789cbb04b06154f403da7d52de9780a968acc6002
V5 selection	ef1b3cbb783364b662974c0a605b42a9758433b2b77c864a644d08f1d6bfa78f
V1 rule	a9f1aef9f00e5bb98aa7fe9e294771f279c0693a9815ae40a1ea34167d469f79
V6 protocol (ES)	6bae2857c9bf6910aa728be2cd65da5f510400190f2e45a1d909194bd48ea195
V6 plan file	feb9d0cbb236831a864a4f8981772262953f198fa53570ecde01b3704c95f766

The frozen causal source used for measurement corresponds to commit:

033450e50d6a13b355b010cabaacf686a741d25f

Its identity remains distinct from the historical reference's source identity. No hash in this report confers scientific admission or training authorization.

10.2 Closing interpretation

The results justify measuring factual contributions alongside activity and aggregate loss. They also document a real limit of the initial design: none of the three candidates is an admissible control under the fixed rule. V6 proposes a different, explicit comparison for studying persistent blocking. The causal hypothesis remains open until that study is executed and evaluated within its declared scope.